



School:Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment: Frontend Connect – Web3.js Integration

***Coding Phase: Pseudo Code / Flow Chart / Algorithm**

- Open Remix IDE and write the SimpleStorage.sol smart contract.
- Compile the contract using the Solidity compiler in Remix.
- Copy the generated ABI (Application Binary Interface) after successful compilation.
- Deploy the contract to the Sepolia Testnet using MetaMask.
- Copy the deployed contract address from Remix.
- Create a React frontend project using create-react-app.
- Add the contract address and network details to a .env file in the React project.
- Install web3.js to enable interaction with the blockchain.
- Connect the frontend to the smart contract using the ABI and contract address.
- Design the user interface in App.js and use web3.js functions to store and retrieve data from the blockchain.

*** Software used:**

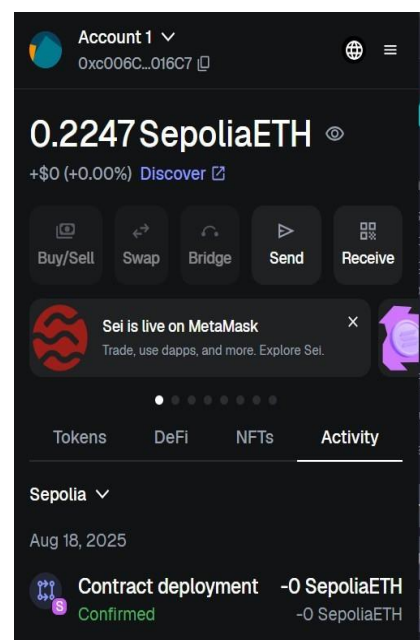
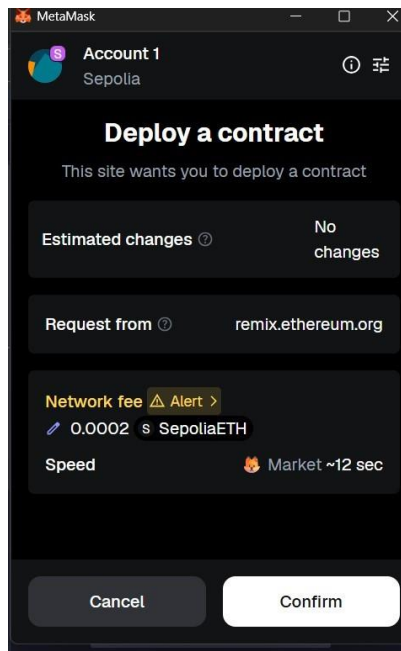
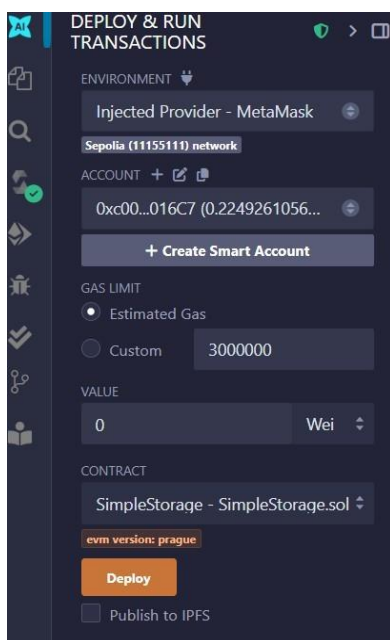
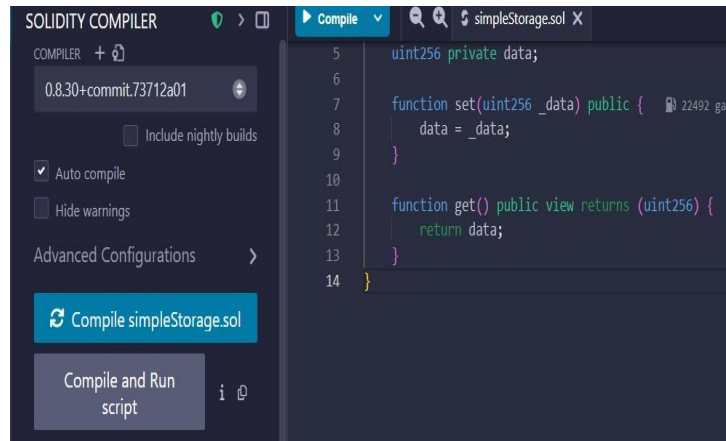
- Laptop
- Visual Studio Code (code editor)
- MetaMask wallet (browser extension)
- Remix IDE (web-based smart contract development tool)
- Node.js
- React (created using create-react-app)
- Web3.js (Ethereum JavaScript interaction library)
- dotenv (for managing environment variables)

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*** As applicable according to the experiment.
Two sheets per experiment (10-20) to be used.**

* Testing Phase: Compilation of Code (error detection)

- Open Remix IDE and create a `.sol` file named `SimpleStorage.sol`, then write the smart contract.
- Compile the smart contract and copy the generated ABI.
- After successful compilation, deploy the contract and select **Injected Provider – MetaMask** as the environment.
- Once deployed, copy the contract address from the **Deployed Contracts** section for later use.
- Use the **web3.js** library to build a frontend interface and interact with the deployed contract through your MetaMask wallet.



* Implementation Phase: Final Output (no error)

- Create a folder named **frontend**, open the terminal, and navigate to the frontend directory.
- Inside the frontend folder, create a **.env** file to store the contract address.
- In the **frontend/src/** directory, create an **ABI.json** file to store the contract ABI.
- Open the **App.js** file and write the frontend code, including the wallet connection function.
- In the terminal, install all required packages using the Node Package Manager (npm).
- Run the project using the command **npm start**.
- Once the app starts, interact with the user interface to connect the wallet and use the **set** and **get** functions.

* Implementation Phase: Final Output (no error)

SimpleStorage DApp (React + Web3)

Connect Wallet

Account:

Enter number

Set Value

Get Value

MetaMask

Welcome back

Unlock

Forgot password?

Need help? Contact MetaMask support

SimpleStorage DApp (React + Web3)

Connect Wallet

Account: 0x19b9a3978978a4165cE5194FDD1CbD4f6a79525F

Enter number

Set Value

Get Value

localhost:3000 says
Connected to MetaMask:
0x19b9a3978978a4165cE5194FDD1CbD4f6a79525F

OK

DEPLOY & RUN TRANSACTIONS

CONTRACT

SimpleStorage - SimpleStorage.sol

evm version: prague

Deploy

Publish to IPFS

At Address

Load contract from Address

Transactions recorded

Deployed Contracts

SIMPLESTORAGE AT 0x238...A

Balance: 0 ETH

set

10

get

Low level interactions

CALLDATA

Transact

MetaMask

Account 1

Sepolia

Transaction request

Estimated changes

No changes

Request from

remix.ethereum.org

Interacting with

Alert

0x2388F...aC7Cd

Network fee

0.0001

SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

Account 1

0xc006C...016C7

0.2247 SepoliaETH

+\$0 (+0.00%)

Discover

Buy/Sell

Swap

Bridge

Send

Receive

Sei is live on MetaMask

Trade, use dapps, and more. Explore Sei.

Tokens

DeFi

NFTs

Activity

Sepolia

Aug 18, 2025

Set

Confirmed

-0 SepoliaETH

-0 SepoliaETH

Contract deployment

-0 SepoliaETH

SimpleStorage DApp (React + Web3)

Connect Wallet

Account: 0x19b9a3978978a4165cE5194FDD1CbD4f6a79525F

10

Set Value

Get Value

Stored Value: 10

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student :

Name :

Regn. No. :

Signature of the Faculty :

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Two sheets per experiment (10-20) to be used***